

# Hui Lin

Machine Learning · Computer Vision · Signal Processing

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## Education

|                                                                                             |                   |               |                                |
|---------------------------------------------------------------------------------------------|-------------------|---------------|--------------------------------|
| <b>Ph.D. in Electrical Engineering</b>                                                      | <b>3.9/4.0</b>    |               | 09.2019 - 05.2025              |
| <i>Northwestern University, advised by Aggelos Katsaggelos &amp; Daniel Kim</i>             |                   |               | <i>Evanston, Illinois, USA</i> |
| <b>M.S. in Mechanical Engineering</b>                                                       | <b>92.7/100.0</b> | <b>rank 1</b> | 09.2016 – 06.2019              |
| <i>Huazhong University of Science and Technology, advised by Bin Li &amp; Xinggang Wang</i> |                   |               | <i>Wuhan, Hubei, China</i>     |
| <b>B.S. in Materials Processing and Control Engineering</b>                                 | <b>90.1/100.0</b> | <b>rank 3</b> | 09.2012 – 06.2016              |
| <i>Huazhong University of Science and Technology (Qiming College)</i>                       |                   |               | <i>Wuhan, Hubei, China</i>     |

## Skills

|                   |                                                                                    |
|-------------------|------------------------------------------------------------------------------------|
| Machine Learning: | CNN, ResNet, RNN, GAN, UNet, Transformer, YOLO, SSD, GNN, Diffusion, LLM           |
| Tools:            | PyTorch, Docker, Git, CUDA, Numpy, Pandas, Opencv, Scikit-learn, SciPy, Caffe, AWS |
| Programming:      | Python, Matlab, SQL, Spark, C++, R, JavaScript                                     |

## Algorithm Competitions

MICCAI 2024 (*FLARE 5th, MyoPS++ 2nd*), ISBI 2024 (*JustRAIGS 5th*), MICCAI 2023 (*ARCADE 3rd*)

## Selected Working and Research Experience (13 projects)

|                                                                                                                                                                                                                                                                                                                                                                                       |                         |                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------|
| <b>Hypertension Detection via Wrist-collected Signals</b>                                                                                                                                                                                                                                                                                                                             | OPPO US Research Center | 06.2024–Present   |
| <ul style="list-style-type: none"><li>Developed models for <b>blood pressure calibration, regression, and classification</b>, feature clustering using <b>PPG</b> and <b>BCG</b> signals following signal preprocessing with band-pass filtering.</li><li>Applied architectures including <b>LLMs, ResNet, Transformer, LSTM</b>, and <b>GRU</b>, and contrastive learning.</li></ul> |                         |                   |
| <b>H5 Game Recommendation System Design</b>                                                                                                                                                                                                                                                                                                                                           | OPPO US Research Center | 02.2025-05.2025   |
| <ul style="list-style-type: none"><li>Developed large-scale feature engineering pipelines with <b>SparkSQL</b> and deployed models such as <b>XGBoost</b> and <b>Wide&amp;Deep</b> to predict CTR, writing ranking results back into <b>HIVE</b> for production use.</li><li>Conduct A/B testing and causal inference analysis.</li></ul>                                             |                         |                   |
| <b>Unsupervised Domain Adaptation for Medical Image Segmentation</b>                                                                                                                                                                                                                                                                                                                  |                         | 06.2023 – 02.2025 |
| <ul style="list-style-type: none"><li>Applied <b>GAN</b> to achieve <b>image generation</b> between modalities without paired data.</li><li>Achieved <b>+11.4% DSC</b> and a <b>+13.1% NSD</b> improvement on large-scale benchmarks.</li></ul>                                                                                                                                       |                         |                   |
| <b>Coronary Artery Segmentation and Stenosis Detection</b>                                                                                                                                                                                                                                                                                                                            |                         | 05.2023 – 02.2024 |
| <ul style="list-style-type: none"><li>Proposed <b>YOLO + UNet</b> ensemble for low-contrast, non-uniform angiography detection.</li><li>Ranked <b>3rd</b> out of <b>200+</b> entries with an F1 score of 0.5348.</li></ul>                                                                                                                                                            |                         |                   |
| <b>Defect Image Sample Generation</b>                                                                                                                                                                                                                                                                                                                                                 |                         | 10.2017 – 06.2019 |
| <ul style="list-style-type: none"><li>Combining <b>CycleGAN + D2GAN</b> for industrial defect synthesis.</li><li>Improved anomaly detection accuracy by <b>+0.80%</b> and defect classification by <b>+2.95%</b>.</li></ul>                                                                                                                                                           |                         |                   |
| <b>LED Chip Defect Detection</b>                                                                                                                                                                                                                                                                                                                                                      |                         | 11.2015 – 06.2019 |
| <ul style="list-style-type: none"><li>Developed a <b>semiconductor inspection system</b> with simultaneous classification and localization in a single CNN.</li><li>Utilized <b>CAM</b> for weakly supervised defect localization.</li><li>Achieved state-of-the-art performance with only <b>5.04% inaccuracy</b>.</li></ul>                                                         |                         |                   |

## Selected Publications (13 First-Author Papers, 1104 citations)

|                                                                                                |                                      |
|------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Defect Image Sample Generation with GAN for Improving Defect Recognition</b>                | IEEE TASE                            |
| <i>Niu, S., Li, B., Wang, X. and Lin, H.</i>                                                   | 335 citations                        |
| <b>Automated Defect Inspection of LED Chip using Deep Convolutional Neural Network</b>         | JIM                                  |
| <i>Lin, H., Li, B., Wang, X. et al.</i>                                                        | 297 citations                        |
| <b>Geometry-agnostic Data-driven Thermal Modeling using GNNs</b>                               | Additive Manufacturing               |
| <i>Mozaffar, M., Liao, S., Lin, H., Ehmann, K. and Cao, J.</i>                                 | 89 citations                         |
| <b>Longitudinal Wrist PPG Analysis for Reliable Hypertension Risk Screening</b>                | ICASSP 2025                          |
| <i>Lin, H., Li, J., et al.</i>                                                                 |                                      |
| <b>DRL-STNet: UDA for Cross-modality Medical Image Segmentation</b>                            | MICCAI 2024                          |
| <i>Lin, H., Schiffers, F., et al.</i>                                                          |                                      |
| <b>Brighteye: Glaucoma Screening with Color Fundus Photographs based on Vision Transformer</b> | ISBI 2024                            |
| <i>Lin, H., Apostolidis, C., Katsaggelos, A.</i>                                               |                                      |
| <b>JustRAIGS: Justified Referral in AI Glaucoma Screening Challenge</b>                        | IEEE Transactions on Medical Imaging |
| <i>Madadi, Yeganeh., Raja, Hina., Lin, H., et al.</i>                                          |                                      |